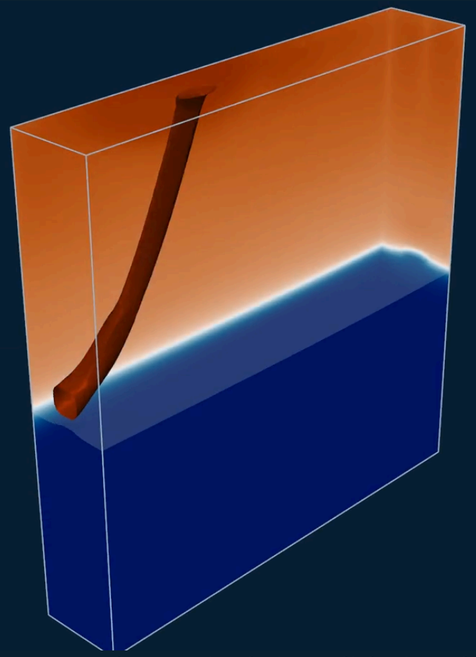
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MASTER'S THESIS DEFENSE · MSC ARTIFICIAL INTELLIGENCE

Can a recorded flow carry a temperature that *pushes back*?

Data-based Numerical Simulation of Buoyant Jet Flow



reference simulation, adiabatic tank, 3D render of stored solver states

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Johannes Kepler University Linz

THE TEST CASE

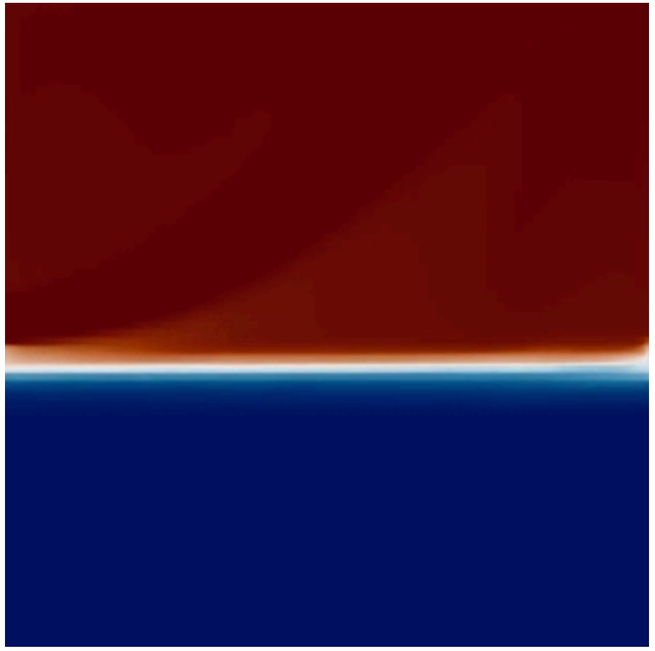
A warm jet builds a front only *five millimetres* thick.

Warm inflow at mid-height for 315 s, then cold inflow until 600 s

Laminar: Reynolds number about 100, no turbulence model

Closed tank, 75 mm: every joule in and out is accounted for

The **outlet** sits just below the front



reference CFD

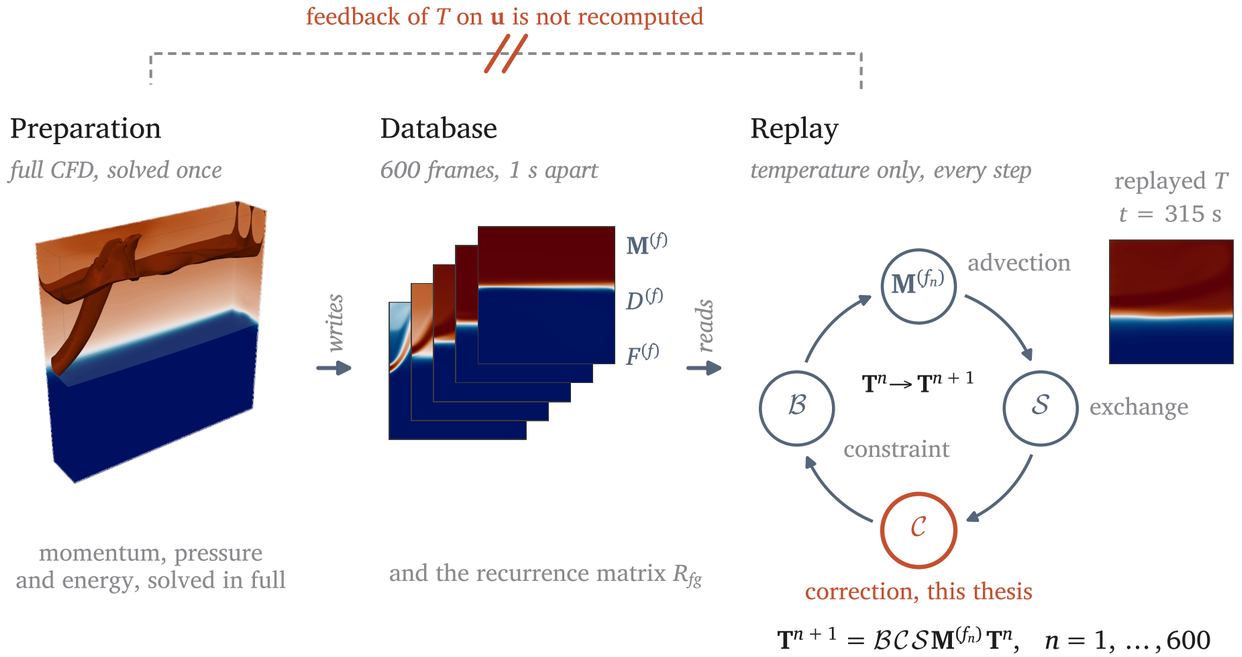
mid-plane temperature, 293-333 K, decoded from the rendered frames (mean error ≤ 0.5 K)

THESIS CH 5

293 K 313 K 333 K

RECURRENCE CFD

Record the motion once, replay it about *260 times* cheaper.



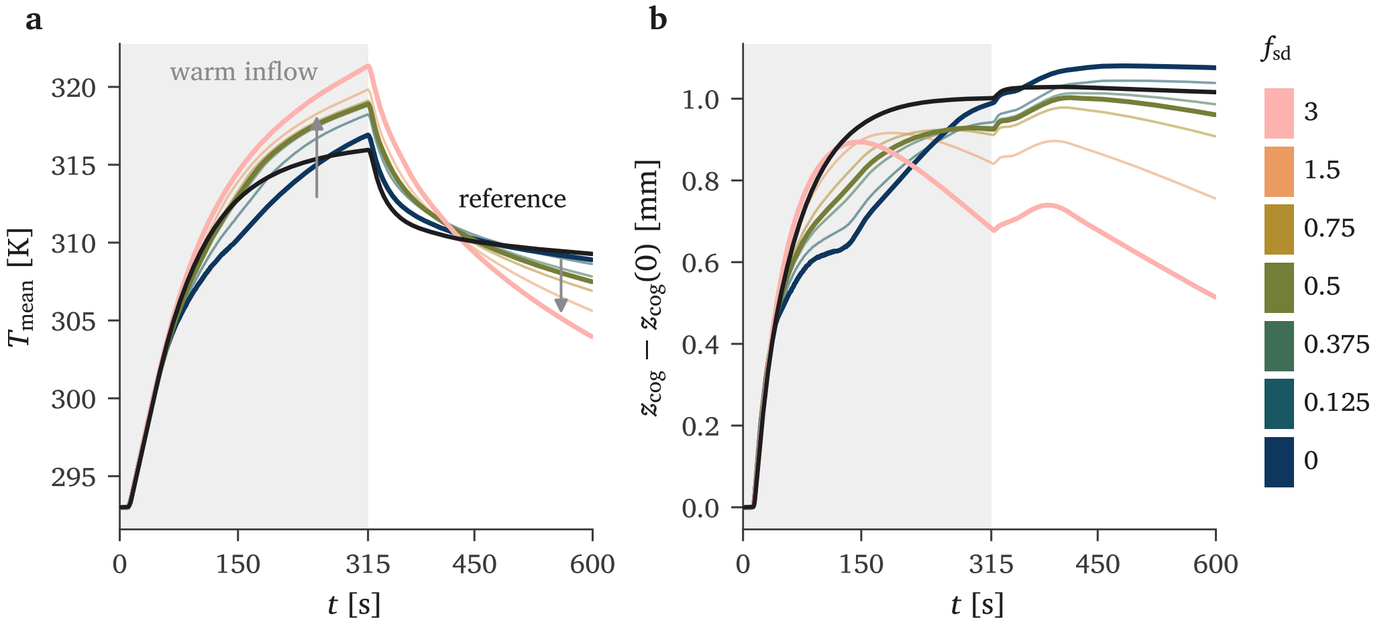


solver time without image output: reference 569 min, mean replay 2.16 min (range 120-700x); the recording itself is one full run

THESIS FIG 2.1 · SEC 5.1

Q1 · THE STANDARD REPLAY

One mixing knob cannot get heat and layer height right *at the same time*.



mean temperature and rise of the centre of gravity, seven strengths of the face exchange

THESIS FIG 5.1

1.07 K

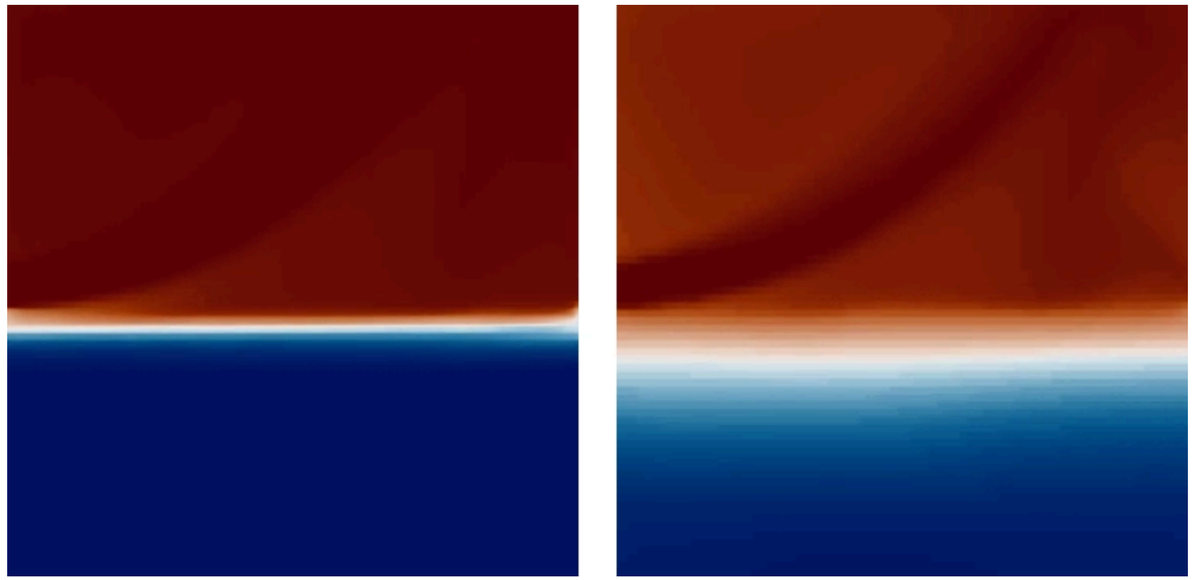
best heat error, with no added mixing; the layer height is best near $f_{sd} = 0.5$

5.08 K

too cold at the end with the strongest mixing

DIAGNOSIS · FROM THE TANK BUDGET DOWN TO ONE FACE

The mixing smears the front, and the outlet below it *drains the heat*.



reference CFD

standard replay, $f_{sd} = 0.5$

mid-plane temperature, 293-333 K, decoded from the rendered frames (mean error ≤ 0.5 K) THESIS CH 5

293 K 313 K 333 K

4.5
mm, reference front at 240 s

19.3
mm, standard replay

4.6 K
of mean temperature drained through the outlet after the inflow

Q2 · THE CORRECTION OF THIS THESIS

A *bounded* anti-diffusion rebuilds the front without new extremes.



model front: 40 K across 60 cells, ν = −0.1 against limited s = 0.25 (not replay data); shaded: outside the data range

THESIS FIG 4.1

Heat moves **uphill**, but both cells stay inside their neighbours' range:

$$\Delta q_f = s \cdot min(Q^+_h, Q^-_c)$$

No new extremes, energy kept face by face, acting across the front.

Limiter in the spirit of flux-corrected transport (Boris & Book 1973; Zalesak 1979).

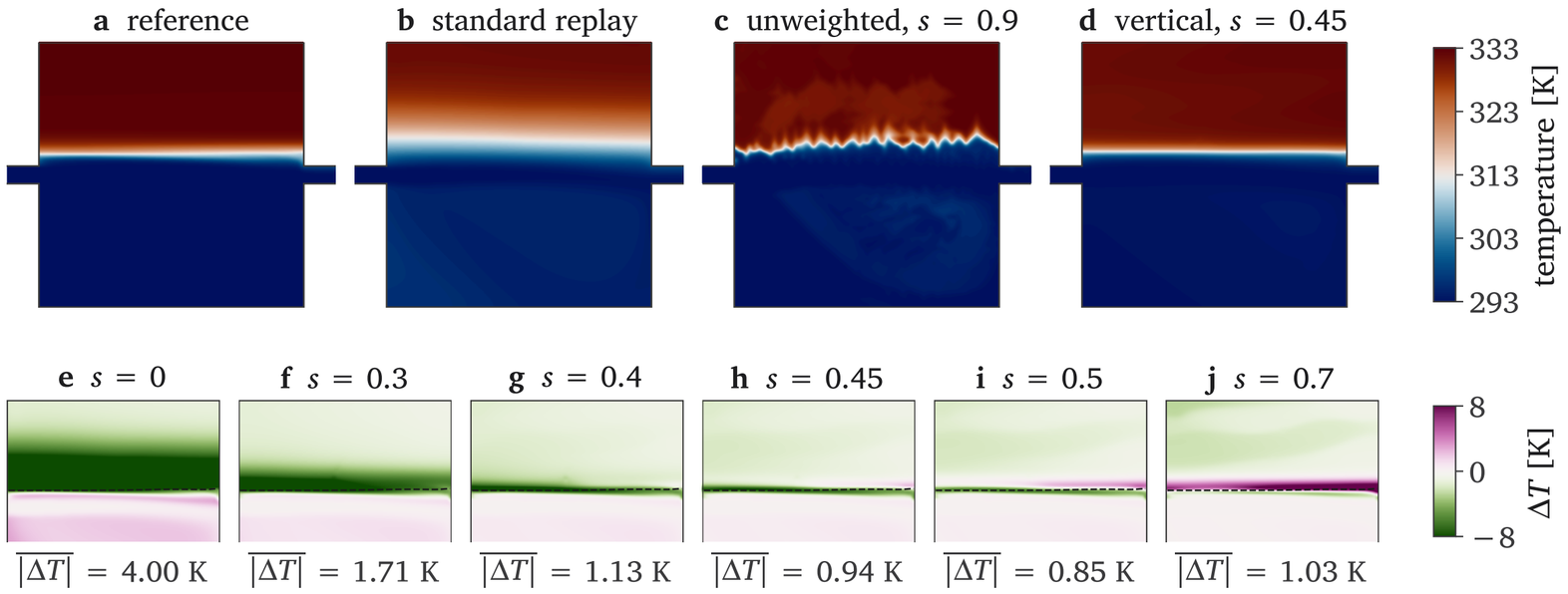
RESULT · INSULATED LID

The front returns to the *width of the reference*.

15.7 mm
standard replay, end front

5.3–4.5 mm
corrected, s = 0.40–0.50; reference 5.0 mm

Heat error over the run: 1.46 → about 0.7 K. Strength chosen against the reference.

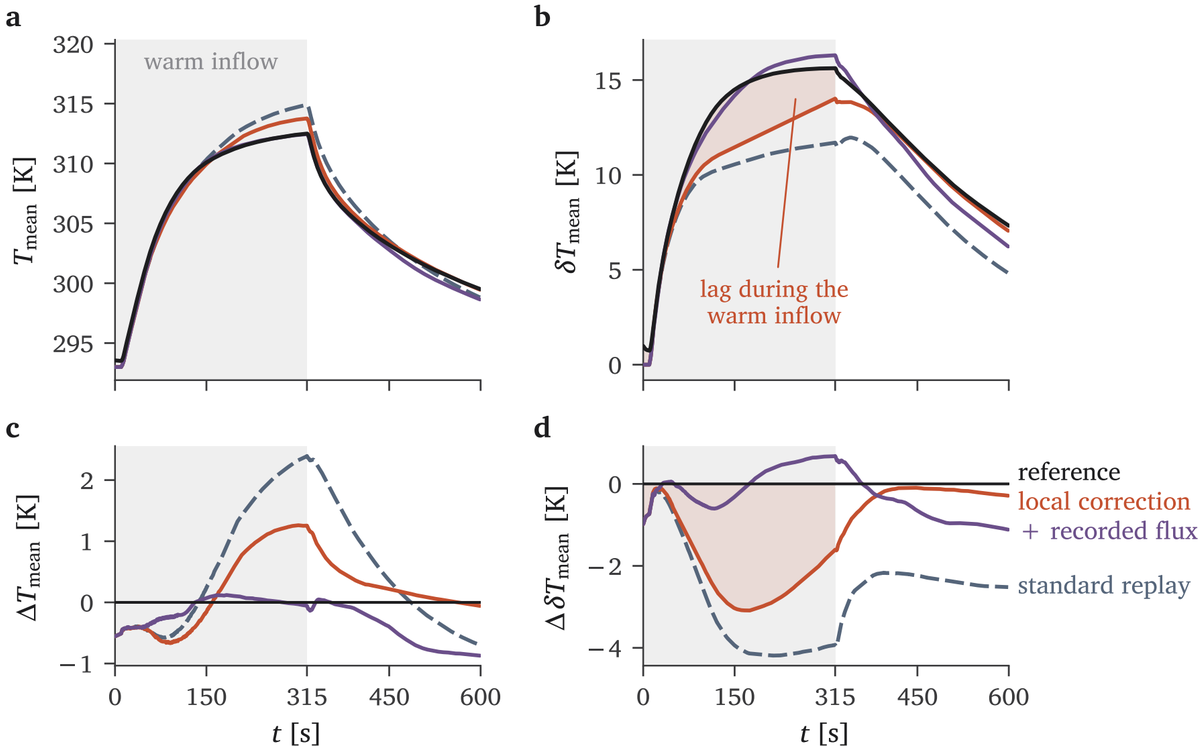


end state at 599 s and the strength ladder as departure from the reference; s was chosen by comparison with this reference

THESIS FIG 5.9

Q3 · Q4 · COOLED LID

With a cold lid the heat comes out right, but the layers *separate too slowly*.



mean temperature, heterogeneity and their departures; band: lag of the local correction while the tank fills

THESIS FIG 6.3

−0.66 → +0.06 K

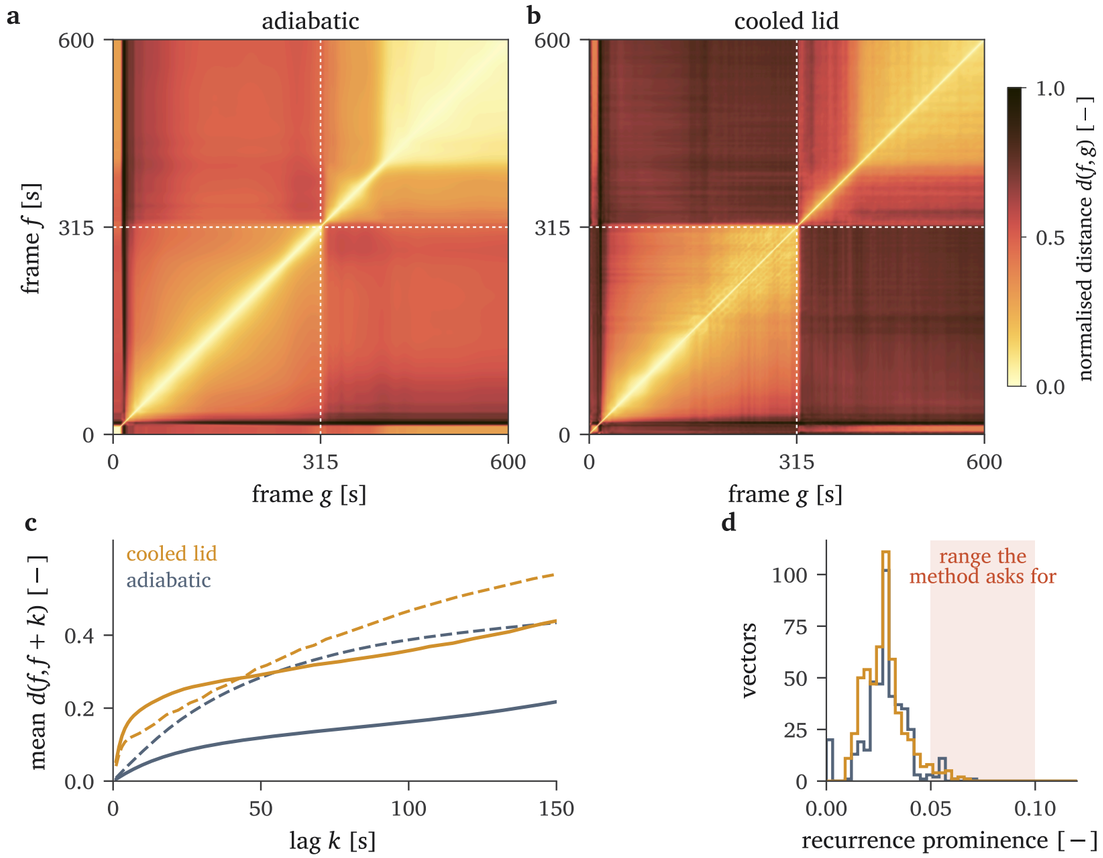
end error on 9.62 K of cooling: standard replay, then the correction at s ≈ 0.55

2.25 → 0.45 K

layering lag while filling, without and with a recorded heat flux; the price: the tank ends 0.86 K too cold

WHY THE RECORDING CANNOT DO IT

A recording belongs to the *flow regime* it was made in.



(c) after the inflow the cooled-lid flow renews about ten times faster at short lags; (d) neither recording recurs in the method's own terms

THESIS FIG 2.2

EVIDENCE AND LIMITS

What the evidence supports, and what it *does not*.

SHOWN

1. no single f_{sd}; the drain at the outlet is the cause
2. a conservative, bounded correction rebuilds the front
3. under cooling, the remaining error is missing transport
4. a recording belongs to its flow regime

OPEN

- OPEN a strength chosen without the reference
- OPEN other meshes, turbulent or multiphase flows
- OPEN the through-flow vessel (Q5): runs in progress
- OPEN a recorded flux that transfers to other cases

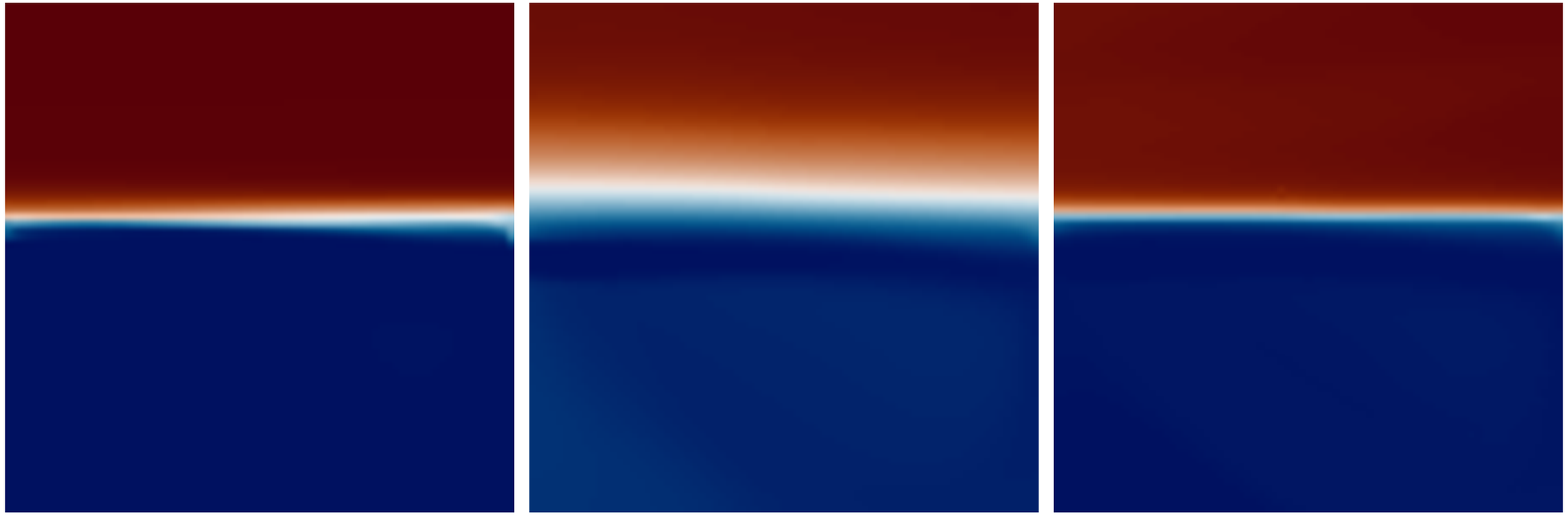
THREE THINGS TO TAKE HOME

1. A measured trade-off in the standard replay, and its cause: a smeared front drains heat through the outlet.
2. A conservative, bounded anti-diffusive correction that rebuilds the front without new extremes.
3. The limit of local corrections: what remains is transport, and a recording belongs to its regime.

Thank you.

BACKUP · FRAME INSPECTION

Stored snapshots: reference, standard and corrected replay.



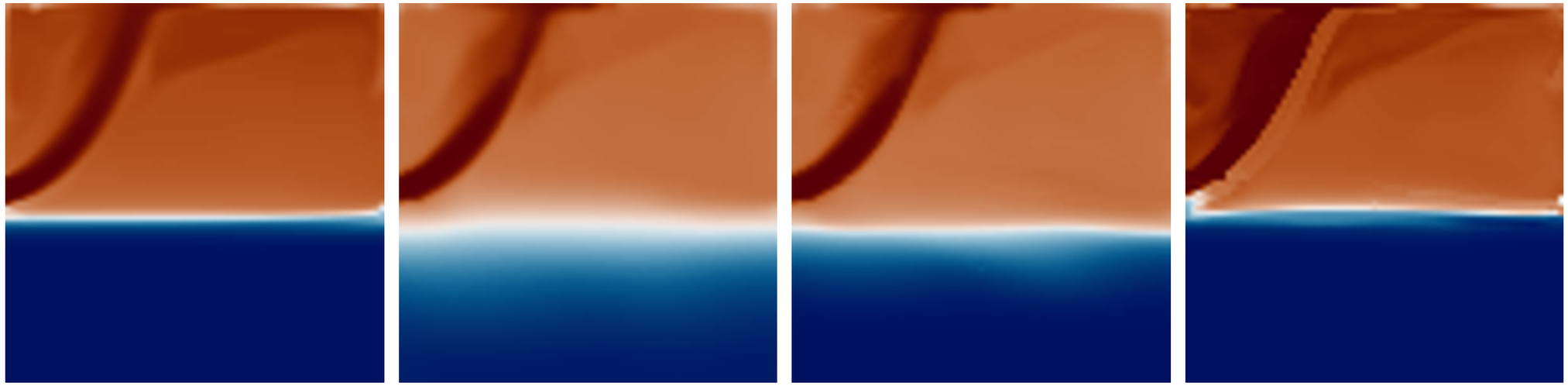
reference CFD | standard replay | corrected, s = 0.45

mid-plane temperature, 293-333 K; solver snapshots of the replays, reference states interpolated in time

STORED SOLVER SNAPSHOTS

BACKUP · FRAME INSPECTION

Cooled lid: the recorded flux restores the layering while the tank fills.



reference CFD | standard replay | corrected, s = 0.475 | + recorded flux

cooled lid, h = 100; same scale 293-333 K

STORED SOLVER SNAPSHOTS

BACKUP · DERIVATION (THESIS APP. D)

Below $s = 1/6$ no cell can leave its bounds, even before the clamp.

Each face gives at most $s \cdot Q^{+}_{h}$ to the warm cell and takes at most $s \cdot Q^{-}_{c}$ from the cold one.

A cell with n_f faces receives at most $n_f s \cdot Q^{+}$ and gives at most $n_f s \cdot Q^{-}$.

So $-Q^{-} \le C \cdot \Delta T \le Q^{+}$ whenever $n_f s \le 1$, i.e. $\mathbf{s \le 1/6}$ on the hexahedral mesh.

Above 1/6 the clamp may act; the working range 0.40–0.50 relies on the clamp and the global conservation constraint.

BACKUP · COST

260 times is solver time against solver time, without the recording.

Reference: 569 min of solver time for 628 s of flow, image output removed

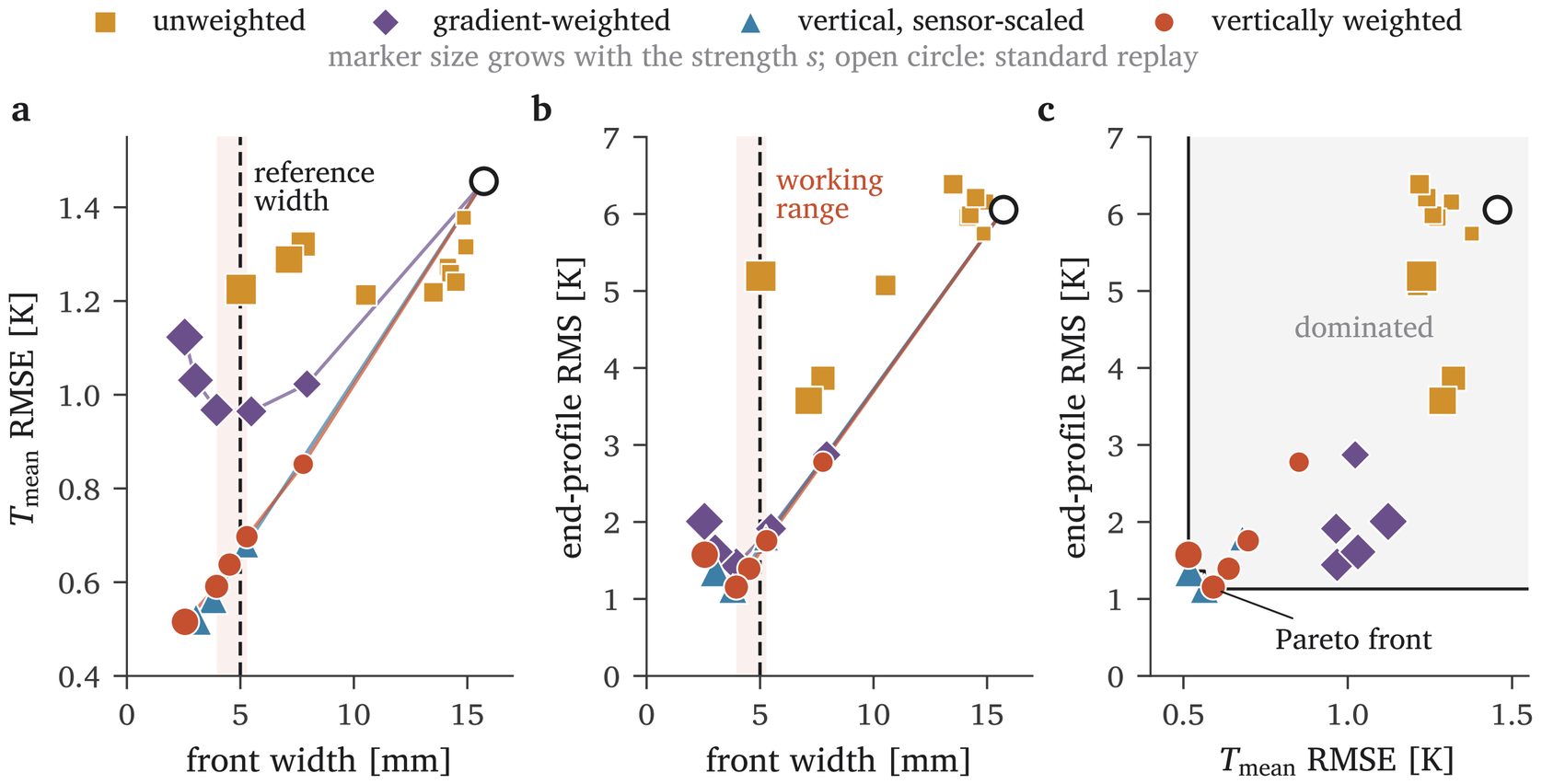
Replay: 0.81–4.73 min, mean 2.16 min for 600 s; grows with f_{sd} (inner sweeps)

Ratio 120–700x across the sweep, about 260x on average, good to about 5 % (different spans)

The recording costs one full CFD run; the saving pays off from the second replay

BACKUP · TRADE-OFF

The vertical form moves straight towards the reference width.

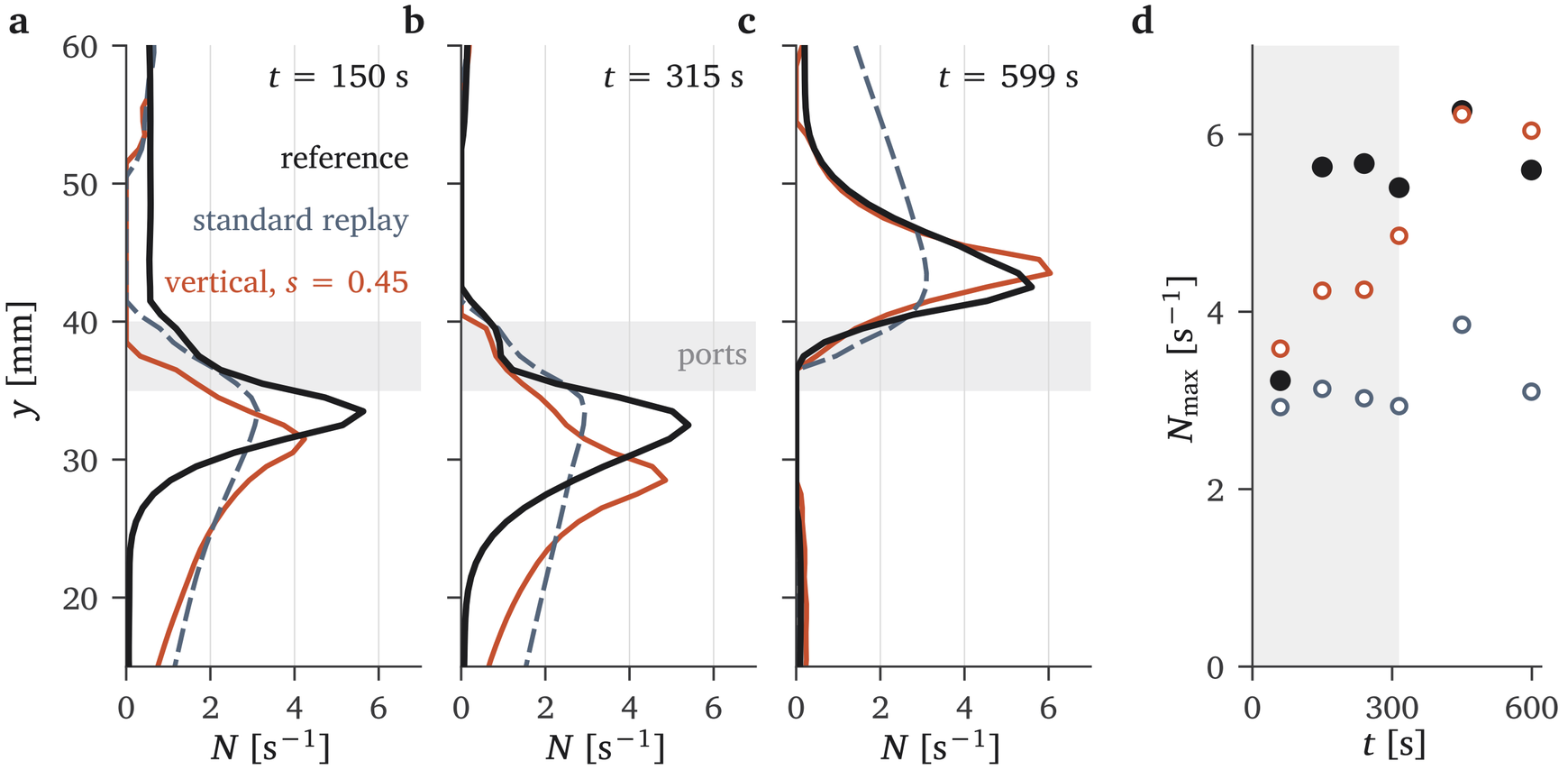


trace error, end profile and front width of every adiabatic correction run

THESIS FIG 5.6

BACKUP · BUOYANCY FREQUENCY

The correction restores the stability of the front, not only its width.

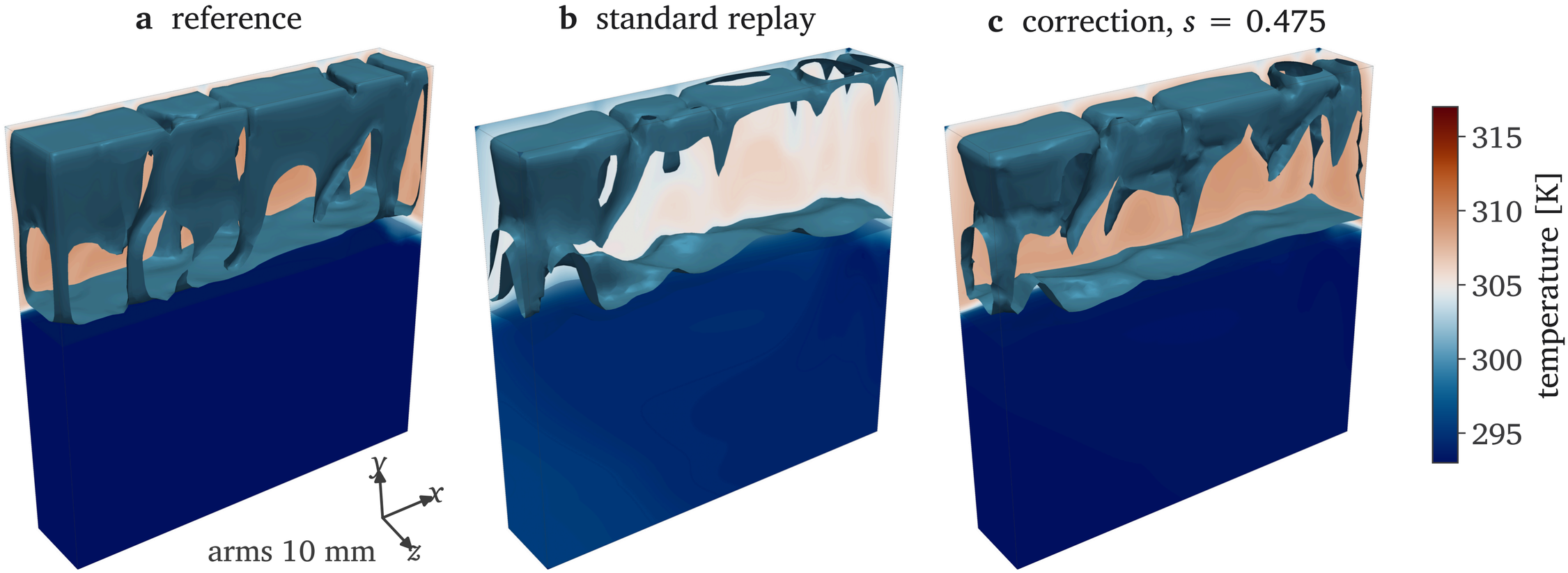


N(y) across the front; peak: reference 5.4–6.3, standard 2.9–3.9, corrected 6.2 and 6.0 s^{-1} at 450 and 599 s

THESIS FIG 5.7

BACKUP · 3D

With the correction the recorded plumes reach the front again.



end state in 3D; the replay invents no motion, the plumes were in the maps

THESIS FIG 6.5

BACKUP · Q5

The vessel study is running; no result is claimed here.

Two replays per feed position (lid-fed, floor-fed), with and without the correction, compared first with their own CFD and only then with thermocouple measurements.